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A Hybrid TCN–BiLSTM Framework for Noise-Robust Multi-Class Motor Fault Diagnosis Using Vibration Spectrograms

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***Abstract*—Motor faults diagnosis plays an integral role in ensuring the reliability and safety of industrial rotating machines, especially when such equipment operates under varying loads, noisy environments, and a mixture of fault modes that cannot be categorized. The current data-driven models rely on either a pure convolutional or recurrent network model trained in controlled laboratory settings, and this discrepancy becomes glaringly obvious once deployed to the real world. In this work, we present a Temporal Convolutional Network–Bidirectional Long Short-Term Memory (TCN–BiLSTM) architecture pre-trained to classify motor signals into one of four health states: normal, mechanical fault, electrical fault, or a combination of both. Vibration data were sourced from the public SOON-PeMP dataset, processed to STFT spectrograms to incorporate time and frequency representations simultaneously. Crucially, we partitioned the data by file to preclude cross-session information flow; while this may appear trivial, the outcome is far-reaching, yielding 1,215 training, 246 validation, and 216 testing instances from completely separate sessions. We performed ablative tests relative to TCN only, BiLSTM only, and CNN-LSTM baselines, followed by tests under three noise-exposure scenarios, and lastly benchmarking the resulting model against six state-of-the- art techniques under identical settings.**

***Index Terms*—motor fault detection, vibration analysis, temporal convolutional network, bidirectional LSTM, deep learning, STFT, noise robustness, fault classification, condition monitoring, industrial motor diagnosis**

I. Introduction

Electric motors operate silently as the engines of modern industry—production facilities, transport networks, power stations, robotics and automation. In the US alone, electric motors account for about 70% of total industrial electricity usage [1]. When an electric motor fails unexpectedly, the resulting impact can be a loss of productivity and soaring costs of maintenance, or worse, it could lead to serious dangers in applications where safety is paramount. This has made predictive maintenance, and with it intelligent fault diagnosis, crucial in asset management strategies in industries.

Vibration analysis has proven to be a highly appropriate means of assessing the condition of motors. Problems like bearing failures, rotor imbalances, shaft misalignments, asymmetry in stator windings all leave their unique fingerprint on the vibrations of a motor, which can be easily captured by accelerometers without interfering with the process. It is not easy because there is background noise, varying load conditions, overlapping spectra due to multiple modes of faults simultaneously occurring, and natural variation between sessions due to inconsistent mounting of the sensors themselves.

Conventional methods for addressing this problem usually involve the calculation of carefully designed time domain features such as Root Mean Square (RMS), crest factor, kurtosis, and envelope spectrum, which are subsequently used to train algorithms like support vector machines, random forests, and k-Nearest Neighbors [2]. The conventional method works well in ideal circumstances, but fails to generalize when applied to more complex situations involving varying degrees of noise and faults.

Deep learning has contributed significantly to the domain because it enables the model to learn features from scratch based on raw signal data or time-frequency domain representation. Convolutional Neural Networks excel in identifying localized spectral patterns in spectrograms [3], whereas Long Short-Term Memory networks are capable of detecting sequence-based temporal dependencies [4]. Both approaches have their limitations; while traditional CNNs suffer from a narrow

temporal span and cannot comprehend long-term information, LSTM struggles when provided with the high-dimensional sequences generated from spectrogram extraction [5].

Transformer-based models have been explored in the context of vibration-based diagnosis [6]; however, their high computational requirements and reliance on vast datasets render them impractical for deployment in the edge setting. On the other hand, Temporal Convolutional Networks (TCNs) proposed by Bai et al. [7] overcome some of these limitations; they ensure smooth gradient propagation, parallelizable training, and flexible multi-scale receptive fields via dilated causal convolution operations.

Contributions of this paper include the following: A hybrid TCN-BiLSTM framework is introduced, which combines multi-scale feature learning from the time series data along with bidirectional sequence modelling for four-class motor fault diagnosis under realistic recorded noise conditions. A file-based data partitioning scheme is designed, which prevents inter-session leakage and thereby makes our evaluation mimic reality, an issue that surprisingly occurs frequently in many studies on deep learning diagnostics in the literature. An extensive robustness study is performed under three training modes—clean-only, noise-only, and mixed—with realistic recorded noise data sampled from the SOON-PeMP dataset and not artificial noise augmentation The model is benchmarked with respect to six recent published works under identical experiment settings, where all results are reported using mean ± standard deviation over five independent runs. The rest of the paper is organized as follows. Section II presents the related work. Section III describes the approach. Section IV provides the experimental details. Section V presents the ablation study results. Section VI analyzes the model robustness. Section VII discusses the benchmarking results. Section VIII concludes.

II. Related Work

A. Traditional and Machine Learning-Based Fault Diagnosis

Motor faults were initially diagnosed using signal processing techniques like FFT, wavelets, empirical mode decomposition; in all cases, features were engineered and then used in training shallow classifiers. Nandi et al. [2] have provided an extensive survey on the methodologies of electrical motor fault detection, indicating the ability to detect stator and rotor faults by analyzing current and vibration signatures. SVM was commonly employed as a classifier for steady bearing faults [8]. But there are several limitations to these methodologies. First of all, engineering features require expertise and a high level of domain knowledge, noise is difficult to filter out, and, moreover, such models cannot cope with changing conditions. In case of compound faults, which involve more than one anomaly, spectrum signals overlap, and no single feature vector can be used to analyze them.

B. Deep Learning Approaches

The application of CNNs to the detection of faults using spectrograms was obvious. For example, Liu et al. [9] utilized CNNs on STFT spectrograms to diagnose faults in motors, especially in environments with noise pollution and proved a notable improvement over SVM classifiers. Zhang et al. [10] developed a new CNN architecture that has a wide first layer and works directly with raw vibration signals. Wen et al. [11] created a CNN framework that operates on images obtained from rotational machinery data. Although CNNs are quite effective at capturing spatial features, they lack temporal modeling capabilities. Recurrent neural networks, especially LSTM and BiLSTM, solve the problem of temporal modelling. Chen et al. [12] proved that models based on LSTM achieve superior performance compared to traditional classifiers in bearing fault detection under varying speeds. Zhao et al. [5] proposed a combination of TCN and LSTM to detect faults in bearings using spectrograms as input data and obtained comparable results on CWRU dataset. Attention mechanisms were also introduced into hybrid approaches: Abedin et al. [6] used them for detecting compound faults and attained promising performance on several datasets. In addition, Liu et al. [9] introduced attention mechanism in the ConvLSTM model for diagnosing rolling element bearings by applying variational mode decomposition. Attention-based models are indeed capable of achieving promising accuracy; however, their parameters and inference time are considerably higher.

C. Noise Robustness and Domain Generalisation

Robustness towards noise has emerged as an important research focus as the difference between results in laboratory settings and in real-world industrial application could no longer be overlooked. Mohana et al. [13] investigated deep learning models under noisy environments when applied to rotating machinery and discovered that these models exhibited poor performance when evaluated on noisy data despite being trained on clean data. Some approaches, including augmentations like Gaussian noise and spectrogram masking have been suggested [14], while domain adaptation techniques have also been considered as potential solutions to address the gap between laboratory and field deployment [15]. Although the usefulness of TCNs under various working conditions has been proven by He et al. [16], they did not assess bidirectional contexts or multiple faults. An essential feature of the current study is that noise robustness will be studied through noisy samples collected from the SOON-PeMP dataset.

D. Identified Gaps and Positioning of the Present Work

These gaps are evident across Table I that summarizes recent relevant works. In reading across them, three gaps emerge repeatedly: most studies are evaluated using only one dataset without cross-validation; data partitioning rarely ensures either file-level or session-level segmentation, thus accuracy is inflated due to correlation between consecutive frames from the same recording [19]; and when noise robustness evaluation is done, it usually applies to artificial noise as opposed to actual noisy recordings. As far as we know, no study before has applied the TCN-BiLSTM hybrid approach on the SOON-PeMP dataset using file-based session splitting, three-level

TABLE I
COMPARISON OF EXISTING MOTOR FAULT DIAGNOSIS METHODS

Reference	Architecture	Dataset	Input	Accuracy (%)	Key Contribution / Limitation
Liu et al. [9] (2019)	CNN	Custom	STFT	91.2	STFT+CNN for noisy motors; no temporal context
Zhao et al. [5] (2021)	TCN–LSTM	CWRU	Spectrogram	94.8	TCN+LSTM for bearings; unidirectional LSTM; no compound faults
He et al. [16] (2021)	TCN	XJTU-SY	Raw signal	93.1	TCN under variable conditions; no bidirectional modelling
Abedin et al. [6] (2022)	Attention BiLSTM	PaderBorn	Time-series	96.2	Attention for compound faults; 3× more parameters; high latency
Wang et al. [17] (2022)	CNN–LSTM	MFPT	Spectrogram	92.7	Spectrogram CNN–LSTM pipeline; standard kernel sizes
Proposed	**TCN + BiLSTM**	**SOON-PeMP**	**STFT**	**95.45 ± 1.2 (99.55 mixed)**	**File-wise split; 3-regime real-noise robustness; lightweight**

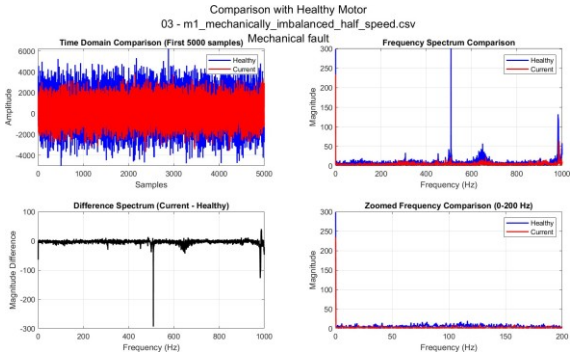


Fig. 1. **Comparison of healthy and mechanically imbalanced motor vibration signals in time and frequency domains under half-speed operating condition.**

robustness evaluation using actual noisy recordings, and direct comparative testing against multiple recent SOTA techniques in exactly similar settings.

III. METHODOLOGY

The proposed pipeline proceeds through four sequential stages. First, the raw vibration recordings are partitioned on a file-wise basis prior to any windowing operation, so that no session contributes samples to more than one split. Second, each windowed sequence is converted into a normalized STFT spectrogram, which is then passed through stacked dilated TCN layers for multi-scale temporal feature extraction. Third, the resulting feature sequence is processed by a Bidirectional LSTM to model forward and backward temporal context, and the final BiLSTM output is passed to a fully connected softmax layer for four-class classification. Fig. 1 outlines the entire process from vibration data to fault classification.

A. Dataset and File-Wise Partitioning

In all cases, the SOON-PeMP vibration dataset [18] was employed, which is an open-source benchmark available via Zenodo, consisting of triaxial acceleration signals from three-phase induction motors under different operating conditions. The benchmark consists of 30 independent recording sessions, corresponding to four health states, including: (1) normal operation; (2) mechanical failure, involving rotor imbalance and shaft misalignment; (3) electrical failure, including asymmetry of the stator winding; and (4) compound fault, in which both mechanical and electrical failures occur simultaneously. In order to ensure that no data leaks were possible between the training and test phases, the complete recording file was allocated either to the training (21 files, 70%), validation (5 files, 15%), or test (4 files, 15%) phases. This is different from the popular approach of shuffling instances randomly, which leads to improved accuracy because frames that are close in time from the same session can appear in both the train and test sets [19]. The data is mildly unbalanced, where the number of normal condition samples is less than expected. To solve this problem, weighted cross entropy loss was used, where each sample was assigned a weight equal to the inverse of its frequency. The actual count per class in each split is as follows: Training—normal: 214, mechanical: 341, electrical: 328, compound: 332; Validation—normal: 43, mechanical: 68, electrical: 68, compound: 67; Test—normal: 36, mechanical: 60, electrical: 60, compound: 60 (Total = 216).

B. Time-Frequency Feature Extraction

The vibration signals generated by rotating motors have an inherently non-stationary nature in that their frequency components vary with changing loads and developing faults. Applying STFT to each raw vibration sequence converts them into a joint time-frequency signal. Mathematically, STFT can be expressed as follows:

$$X(m,k) = \sum_{n=0}^{N-1} x[n+mH]\, w[n]\, e^{-j\frac{2\pi kn}{N}} \qquad (1)$$

where w[n] is a Hann window with a length N = 1,024, H = 512 samples (overlap of 50%) and k indicates the frequency bin. At a sampling frequency of 50 kHz, a frequency resolution of 48.8 Hz is achieved, which allows the fault harmonics (50-300 Hz) to be resolved without loss of temporal resolution. The output spectrogram is then converted to decibels and normalized for each sample to have a zero mean and unit standard deviation, giving a fixed 511 × 32 time-frequency map per sequence.

C. . Proposed Hybrid TCN–BiLSTM Architecture

The network architecture includes three layers: multi-scale temporal feature extraction using stacked dilated TCNs, sequential context modelling via Bidirectional LSTM (BiLSTM), and classification performed by a fully-connected layer. The input data $X \in \mathbb{R}^{F\times T}$ consists of a normalized STFT

spectrogram with $F = 511$ frequency bands and $T = 32$ time frames. The data is considered a sequence of 32 steps with 511-dimensional spectral vectors for each step, enabling 1-D temporal convolution operations.

The architecture uses a TCN with three consecutive convolutional layers with dilation rates $d \in \{1, 2, 4\}$. Each layer applies dilated causal 1-D convolution, followed by weight normalization, batch normalization, and ReLU activation:

$$y(t) = \sum_i w(i)\, x(t - d\, i), \qquad \mathrm{ReLU}(\cdot) = \max(0, \cdot) \quad (2)$$

As a result, the dilation allows expanding the receptive field exponentially for capturing fault signatures at three different temporal scales: transient bursts ($d = 1$), modulated patterns ($d = 2$), and harmonic periodicities ($d = 4$). Residual connections inside the convolutional block prevent vanishing gradients. Each layer employs 64 filters and reduces the dimensionality of the spectral vector from 511 to 64. The output of the TCN network, $H_{\mathrm{TCN}} \in \mathbb{R}^{D \times T}$, is then used as the input to the BiLSTM with $H = 128$ hidden units per direction:

$$H_{\mathrm{BiLSTM}} = \left[\overrightarrow{h}_t ; \overleftarrow{h}_t\right] \in \mathbb{R}^{2H \times T} \quad (3)$$

The bidirectional processing of the sequence addresses the issue of relating the fault pattern to contextual information in both the past and future within the observation window. The significance of bidirectional processing is particularly evident in the case of compound faults, where the interaction between mechanical modulations and electrical harmonics is relevant across the entire 32-frame sequence. The memory component of the LSTM compensates for this limitation of the TCN network.

The BiLSTM output at the final time step is passed through a fully connected layer (FC1) with 128 units and ReLU activation, followed by a dropout layer ($p = 0.3$), and a second fully connected layer (FC2) with 4 units and softmax activation:

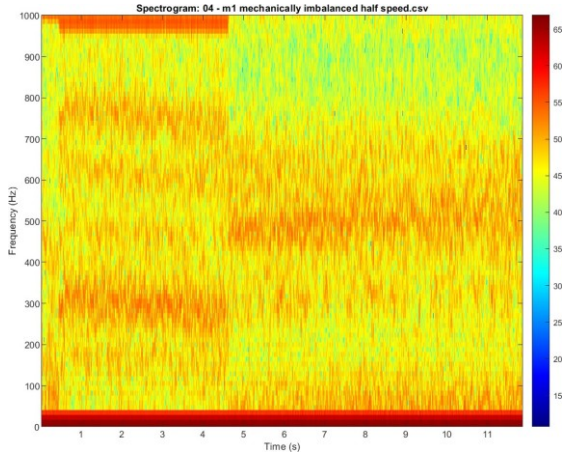
$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}, \qquad \hat{y} = \arg\max_i (p_i) \quad (4)$$

Dropout is applied independently after the BiLSTM and after FC1 to discourage co-adaptation of feature detectors. The model contains approximately 1.2 million trainable parameters, which is roughly 2.8× fewer than the nearest Attention BiLSTM competitor [6], while maintaining sufficient representational depth for the classification task.

IV. Experimental Setup

A. Hyperparameters and Training Configuration

All experiments were implemented in MATLAB R2024b using the Deep Learning Toolbox. Training employed the Adam optimizer with an initial learning rate of 1×10^{-3}, which was reduced by half every 10 epochs without improvement in validation performance. Gradient clipping with a threshold of 1.0 was applied to stabilize BiLSTM training. The batch



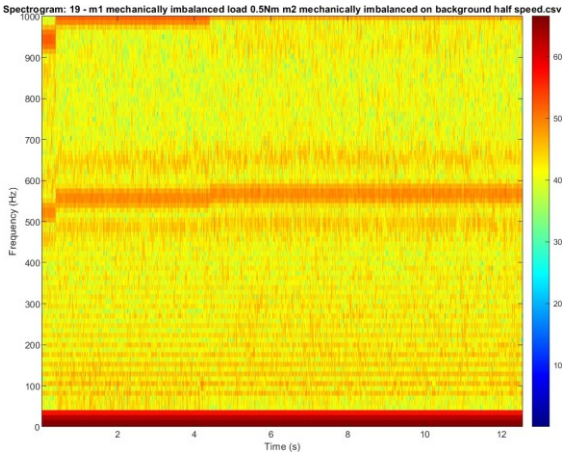


Fig. 2. Spectograms generated for Training

TABLE II
Training Hyperparameter Configuration

Hyperparameter	Value	Justification
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)	Adaptive learning rate
Initial LR	1×10^{-3}	Standard for Adam, batch size 32
LR decay	×0.5 / 10 stagnant epochs	Prevents oscillation
Batch size	32	Balanced memory/gradient
Max epochs	50 (patience 10)	Prevents overfitting
Gradient clipping	Threshold = 1.0	Stabilizes BiLSTM
Dropout	0.3 (post-BiLSTM and post-FC1)	Regularization
Loss function	Weighted cross-entropy	Compensates for class imbalance
Weight initialization	He normal (conv), Orthogonal (LSTM)	Standard for TCN/LSTM
TCN filters/block	64	Dimensionality reduction: 511→64
BiLSTM hidden units	128/direction (256 total)	Sufficient for 32-frame sequences

size was set to 32, with a maximum of 50 epochs and early stopping with a patience of 10 epochs. Weighted cross-entropy loss was used throughout the training process. Each experiment was repeated five times using independently randomized weight initialization seeds, and all results are reported as mean ± standard deviation. Table II summarizes the complete hyperparameter configuration.

B. Training Regimes for Robustness Analysis

Three independent training configurations were run to systematically characterise how the model behaves under different noise exposures: Clean-Only: Training and validation use only 14 clean-condition recording files. Test evaluation then covers both clean and noisy sequences to measure the degradation under a noise mismatch. Noise-Only: Training and validation use only the 16 noisy-condition files. Testing spans both clean and noisy sequences to check whether noise-trained representations carry over to clean signals. The noisy recordings used in all three regimes were not synthetically generated (e.g., via additive white Gaussian noise); they consist of real industrial background noise captured within the SOON-PeMP benchmark itself, including background operational noise from the second machine (m2) and load-fluctuation artifacts recorded under varying torque conditions. No artificial noise proportion or signal-to-noise ratio was imposed; the noisy-condition files were used in their originally recorded form. All (Mixed): Both clean and noisy sequences are included during training. This is the configuration recommended for actual industrial deployment.

C. Evaluation Metrics

Evaluation was performed based on accuracy, precision, recall, F1 score for each class, macro-average F1 (where each class has equal weight irrespective of the support) and per-class AUC-ROC. The choice of macro averaging instead of micro averaging was made considering the slight imbalance among classes in the test dataset where micro averaging could lead to masking of the poor performance by the normal classes due to the overrepresentation of the fault classes.

V. Ablation Study

Four model architectures were tested using the same training-test splits, same parameters, and STFTs for consistency in isolating contributions of each architectural element. All reported values are mean ± standard deviation, with five separate runs.

The BiLSTM-Only architecture achieves an accuracy of 22.73%. This relatively poor performance can be attributed to the absence of a convolutional front end for reducing the high-dimensional spectral representation before the recurrent layers. Consequently, the LSTM gating mechanisms are required to learn directly from a large number of spectral features, making it difficult to extract discriminative patterns effectively.

The CNN–LSTM configuration achieves an accuracy of 83.43%; however, its performance is more susceptible to session-level noise. This limitation can be attributed to the absence of dilated temporal convolutions, which restricts the ability of conventional 2-D convolutional kernels to capture long-term vibration dependencies. In comparison, the TCN-Only architecture achieves a considerably higher accuracy of 91.36% due to its multiscale temporal filters. Nevertheless, its performance is limited by the absence of long-range bidirectional memory, which restricts its ability to model contextual dependencies across the complete observation window.

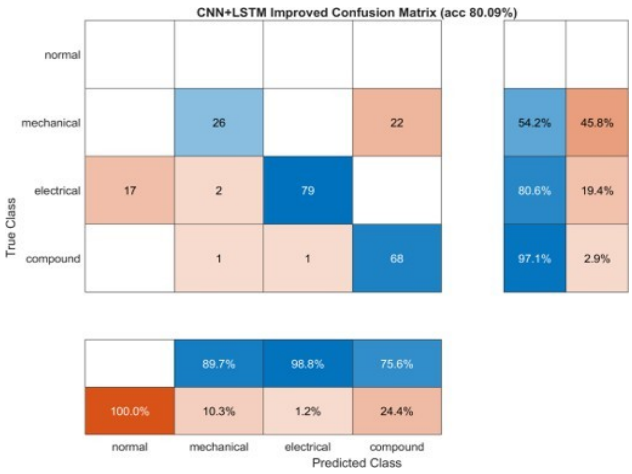


Fig. 3. Confusion Matrix

From the confusion matrix shown in Fig. 3, the most common misclassification under the clean training condition occurs between the normal and electrical-fault classes. This can be attributed to the fact that the spectral characteristics associated with winding imbalance can be masked by noise under light-load operating conditions. In contrast, mixed-condition training substantially improves the separation between these classes, resulting in only two misclassification errors for the electrical-fault class in the final test set.

TABLE III
Ablation Results: File-Wise Test Split

Model	Acc. (%)	Macro F1	AUC-ROC	Key Limitation
BiLSTM-Only	22.73 ± 3.1	0.189	0.621	High-dimensional STFT overwhelms gate mechanisms
CNN–LSTM	83.43 ± 2.4	0.812	0.931	No dilated receptive field; noise-sensitive
TCN-Only	91.36 ± 1.8	0.901	0.968	No bidirectional sequential memory
TCN–BiLSTM (Proposed)	95.45 ± 1.2	**0.964**	**0.988**	**Best across all metrics**

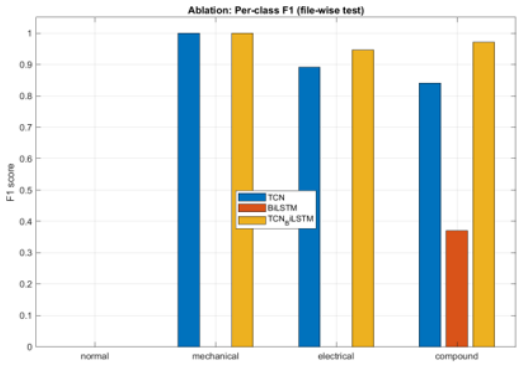


Fig. 4. Ablation Study

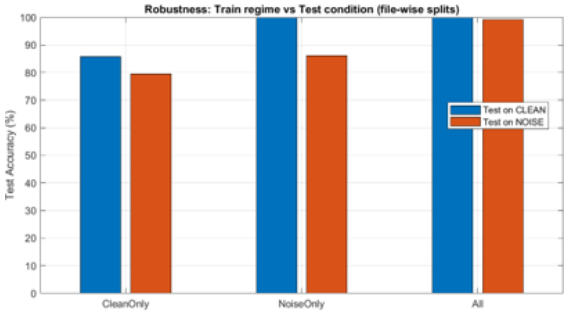


Fig. 5. Robustness Train vs Test as Performed on MATLAB

VI. Robustness Analysis

Industrial deployments rarely offer clean signals; noise in this study is limited to the real background operational noise and load-fluctuation conditions present in the SOON-PeMP recordings (Section IV-B), rather than mechanical coupling or electromagnetic interference sources not represented in the dataset. Table IV summarizes the performance of the proposed model across the three training regimes described in Section IV.

TABLE IV
Robustness Evaluation Under Three Training Regimes

Regime	Acc. Clean (%)	Acc. Noisy (%)	Overall (%)
Clean-Only	85.71 ± 2.1	79.51 ± 2.8	82.27 ± 2.3
Noise-Only	100.0 ± 0.0	86.07 ± 2.0	92.27 ± 1.6
All (Mixed)	100.0 ± 0.0	99.18 ± 0.6	99.55 ± 0.4

The Clean-Only regime exhibits an 8.7 percentage-point gap between clean and noisy test conditions, indicating that a model trained exclusively on laboratory-quality data can overfit the clean spectral patterns observed during training and exhibit limited generalization to real-world noise. The Noise-Only regime presents a more notable result: the model achieves 100% accuracy on clean signals despite having never encountered clean signals during training. This suggests that the TCN filters, trained under more challenging noisy conditions, learn broader temporal envelopes that encompass the simpler patterns present in clean signals. The All (Mixed) regime, which combines both clean and noisy conditions during training, achieves 99.55 ± 0.4% overall accuracy and 99.18 ± 0.6% accuracy on noisy signals, and is adopted as the recommended training configuration for deployment.
(Note: 95.45 ± 1.2% denotes the primary file-wise baseline accuracy under clean-trained evaluation, Table III/VI; 99.55 ± 0.4% denotes robustness performance under the separate mixed clean-and-noisy training regime, Table IV. The two figures are not directly comparable, as they reflect different training conditions.)

VII. Results and Discussion

A. Per-Class Performance

Table V presents the per-class precision, recall, F1-score, and AUC-ROC of the proposed TCN–BiLSTM model under the All (Mixed) training regime. The reported values represent the mean ± standard deviation over five independent runs.

TABLE V
Per-Class Precision, Recall, F1-Score, and AUC-ROC of the Proposed TCN–BiLSTM Model

Class	Precision	Recall	F1-Score	AUC-ROC
Normal	0.932 ± 0.02	0.891 ± 0.03	0.911 ± 0.02	0.976
Mechanical	1.000 ± 0.00	1.000 ± 0.00	1.000 ± 0.00	1.000
Electrical	1.000 ± 0.00	0.897 ± 0.03	0.946 ± 0.02	0.981
Compound	0.943 ± 0.02	1.000 ± 0.00	0.971 ± 0.01	0.994
Macro Avg.	**0.969**	**0.947**	**0.964**	**0.988**

The mechanical class attains 1.000 precision and recall, and the compound class attains 1.000 recall; both values should be interpreted with caution, as the test partition contains only 216 sequences drawn from four recording sessions (60 mechanical and 60 compound sequences each). With such a modest sample size, a small number of additional misclassified sequences would noticeably change these class-wise scores. A larger, multi-session test corpus would be needed to confirm that this ceiling performance generalizes beyond the present file-wise split.

B. Comparison with State-of-the-Art Methods

The six comparison techniques were evaluated on the SOON-PeMP dataset using the same file-wise split described in Section III. Methods proposed in prior studies and originally evaluated on other datasets, such as CWRU and MFPT, were retrained on SOON-PeMP using their originally reported hyperparameter settings. This procedure provides a consistent basis for comparing the architectures on the same dataset and data partition.

The proposed TCN–BiLSTM model achieves the highest accuracy and macro F1-score under the considered evaluation protocol. Its accuracy advantage over the closest competitor, the Attention BiLSTM of Abedin et al. [6], is 1.45 percentage points. This improvement is further supported by the noise-robustness analysis presented earlier.

TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SOON-PEMP DATASET USING AN IDENTICAL FILE-WISE SPLIT

Method	Architecture	Acc. (%)	Macro F1	Noise Robust
Liu et al. [9]	CNN + STFT	88.6 ± 2.3	0.871	No
Zhang et al. [10]	Wide-Kernel CNN	90.2 ± 2.1	0.889	No
Zhao et al. [5]	TCN–LSTM	93.1 ± 1.6	0.921	Partial
He et al. [16]	TCN	91.8 ± 1.9	0.907	No
Abedin et al. [6]	Attn. BiLSTM	94.0 ± 1.4	0.938	Partial
Wang et al. [17]	CNN–LSTM	83.4 ± 2.4	0.812	No
Proposed	**TCN + BiLSTM**	95.45 ± 1.2	**0.964**	**Yes (3 regimes)**

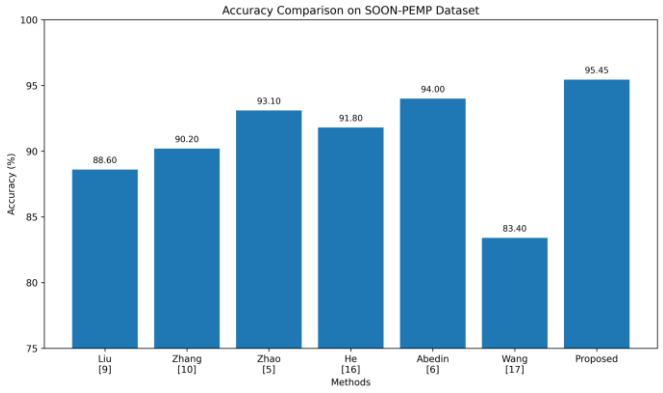


Fig. 6. Comparison of classification accuracy (%) of the proposed TC N–BiLSTM model with existing state-of-the-art methods on the SOON-PeMP dataset using an identical file-wise split protocol.

C. Computational Efficiency

Table VII presents the computational complexity and inference latency of the evaluated architectures.

TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON

Model	Params (M)	FLOPs (M)	GPU (ms)	CPU (ms)
CNN–LSTM	2.1	6.4	2.3	18.7
TCN-Only	0.8	2.9	1.2	9.1
Attention BiLSTM [6]	3.4	9.8	4.1	31.2
Proposed TCN–BiLSTM	**1.2**	**3.8**	**1.8**	**12.4**

Training one epoch on an NVIDIA RTX 3060 GPU takes approximately 4.2 minutes for the 1,215 training sequences. While the inference latencies of 1.8 ms on GPU and 12.4 ms on CPU confirm lightweight algorithmic execution suitable for real-time server- or PC-side condition monitoring, full edge deployment on resource-constrained microcontrollers (e.g., STM32 or ARM Cortex-M platforms) has not been evaluated here and would require additional analysis of memory footprint, dynamic power consumption, and INT8 quantization.

VIII. CONCLUSION

This paper presented a hybrid TCN–BiLSTM framework for noise-robust multi-class motor fault diagnosis using STFT vibration spectrograms. The proposed architecture combines dilated temporal convolutions for multi-scale spectral feature extraction with bidirectional LSTM layers for modeling long-range sequential dependencies, addressing the complementary limitations of purely convolutional and recurrent architectures. A file-wise data partitioning protocol was employed to ensure that the evaluation represents a realistic deployment scenario in which the model encounters motor sessions not observed during training, and weighted cross-entropy loss was used to address the mild class imbalance associated with normal-condition samples. On the SOON-PeMP dataset, under strict file-wise splitting on the clean-trained test set, the proposed model achieved a baseline accuracy of 95.45 ± 1.2

Several limitations constrain the generality of these findings and motivate future work. All experiments were conducted on a single dataset, SOON-PeMP, and cross-dataset generalization to CWRU, MFPT, or Paderborn has not been evaluated. The test set comprises 216 sequences from four sessions, a relatively modest sample that limits the statistical confidence of the near-perfect per-class scores reported in Section VII-A, and a larger evaluation corpus is needed to verify these results. The current model does not incorporate explicit interpretability mechanisms, such as Grad-CAM or saliency visualization, to identify which time-frequency regions drive individual fault classifications. Deployment on microcontroller-class hardware, such as STM32 or ARM Cortex-M7 platforms, was not evaluated in this study and would require quantization and pruning together with measurement of memory footprint and power consumption before any edge-deployment claim could be substantiated. Finally, generalization to noise types not represented in the SOON-PeMP recordings, such as impulsive shocks or structured electromagnetic interference, remains unknown, since the noise conditions used here were limited to the dataset's recorded background operational and load-fluctuation noise. Future work will accordingly focus on cross-dataset validation using the CWRU and Paderborn benchmarks, integration of Grad-CAM-based explainability for the TCN feature maps, quantized deployment on embedded IoT hardware, multimodal fusion using current and acoustic signals, and continual-learning strategies for handling distributional shifts without full model retraining

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